CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 1: Empirical Complexity of Merge Sort vs Quick Sort

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Implement and plot CPU runtime curves of Merge Sort and Quick Sort across varying input sizes.

2. Practical Tasks & Verification Steps:

1. Implement Merge Sort dividing array at midpoint and merging two sorted subarrays in O n space.
2. Implement Quick Sort with randomized pivot selection partitioning array in-place.
3. Generate random test arrays of sizes n = 1,000, 5,000, 10,000, 50,000, and 100,000 integers.
4. Measure execution time using System.nanoTime and plot comparative empirical growth curves.

Submission Guidelines & Verification:

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